

## DERIDDER BEAUREGARD, LA, USA

WMO: 722334

Lat: 30.833N

Lon: 93.333W

Elev: 62

StdP: 100.58

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 03976

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.0       | -0.2 | -9.3                          | 1.7 | 4.0  | -6.4 | 2.2 | 5.2  | 8.8                   | 16.4 | 8.1 | 15.0 | 1.6                   | 0    | 0.353 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 10.1                      | 35.4            | 24.7 | 34.5 | 24.5 | 33.2 | 24.4 | 27.3                | 30.3 | 26.8 | 29.9 | 26.3 | 29.5 | 2.6                     | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.6                            | 22.3 | 28.1 | 26.2 | 21.7 | 27.9 | 25.5 | 20.9 | 27.6 | 86.1          | 30.1 | 84.4 | 30.8 | 82.1 | 29.5 | 29.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| 8.0               | 7.0  | 5.7 | DB                         | -5.0 | 36.9               | 2.5 | 1.9                                                | -6.8 | 38.2       | -8.3 | 39.3       | -9.7  | 40.3       | -11.5 | 41.7 | (4) |
|                   |      |     | WB                         | -5.9 | 27.6               | 2.2 | 1.2                                                | -7.5 | 28.4       | -8.8 | 29.1       | -10.0 | 29.8       | -11.6 | 30.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 19.7  | 9.9   | 12.2  | 15.7  | 19.4  | 23.5  | 27.0  | 27.7  | 27.9  | 25.6  | 20.4  | 14.9  | 11.4 | (6)  |
|      |                                                                   | DBStd       | 7.58  | 5.23  | 5.56  | 4.94  | 3.78  | 3.14  | 1.93  | 1.73  | 1.72  | 2.47  | 4.38  | 5.11  | 5.29 | (7)  |
|      |                                                                   | HDD10.0     | 175   | 66    | 37    | 12    | 1     | 0     | 0     | 0     | 0     | 1     | 14    | 46    |      | (8)  |
|      |                                                                   | HDD18.3     | 976   | 266   | 181   | 109   | 33    | 5     | 0     | 0     | 0     | 31    | 127   | 225   |      | (9)  |
|      |                                                                   | CDD10.0     | 3704  | 63    | 99    | 188   | 282   | 418   | 511   | 548   | 556   | 469   | 322   | 161   | 89   | (10) |
|      |                                                                   | CDD18.3     | 1464  | 4     | 11    | 28    | 64    | 164   | 261   | 289   | 297   | 219   | 93    | 23    | 11   | (11) |
|      |                                                                   | CDH23.3     | 13537 | 5     | 37    | 136   | 447   | 1364  | 2637  | 2966  | 3072  | 1912  | 805   | 125   | 31   | (12) |
|      |                                                                   | CDH26.7     | 5650  | 0     | 3     | 12    | 72    | 474   | 1221  | 1354  | 1451  | 804   | 244   | 14    | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 2.0   | 2.3   | 2.6   | 2.6   | 2.7   | 2.2   | 1.6   | 1.4   | 1.3   | 1.4   | 1.6   | 1.9   | 2.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 1527  | 144   | 111   | 117   | 121   | 119   | 124   | 123   | 100   | 125   | 128   | 137   | 148  | (15) |
|      |                                                                   | PrecMax     | 2032  | 325   | 283   | 209   | 272   | 293   | 315   | 237   | 161   | 336   | 512   | 347   | 303  | (16) |
|      |                                                                   | PrecMin     | 1126  | 22    | 20    | 39    | 14    | 1     | 30    | 38    | 7     | 32    | 0     | 16    | 65   | (17) |
|      |                                                                   | PrecStd     | 262   | 84    | 70    | 49    | 65    | 77    | 71    | 53    | 37    | 63    | 126   | 88    | 59   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 24.0  | 26.9  | 28.0  | 30.2  | 32.9  | 37.0  | 36.3  | 37.4  | 35.2  | 32.4  | 28.6  | 26.0  |      | (19) |
|      |                                                                   | 0.4% MCWB   | 18.0  | 18.8  | 19.2  | 21.3  | 23.1  | 25.1  | 25.3  | 23.5  | 23.1  | 22.5  | 22.2  | 21.3  |      | (20) |
|      |                                                                   | 2% DB       | 22.3  | 24.5  | 26.3  | 28.4  | 32.1  | 35.0  | 35.1  | 35.4  | 33.7  | 31.0  | 26.4  | 24.0  |      | (21) |
|      |                                                                   | 2% MCWB     | 17.7  | 18.8  | 19.1  | 20.7  | 22.8  | 25.1  | 25.2  | 24.4  | 23.3  | 22.5  | 20.5  | 20.9  |      | (22) |
|      |                                                                   | 5% DB       | 20.8  | 22.6  | 25.0  | 27.4  | 30.9  | 33.2  | 33.9  | 34.3  | 32.4  | 29.6  | 24.9  | 22.3  |      | (23) |
|      |                                                                   | 5% MCWB     | 17.7  | 19.0  | 18.6  | 20.0  | 22.6  | 24.8  | 24.8  | 24.3  | 23.7  | 22.0  | 20.2  | 20.1  |      | (24) |
|      |                                                                   | 10% DB      | 18.8  | 21.1  | 23.0  | 26.1  | 29.8  | 32.4  | 32.7  | 32.9  | 31.2  | 27.7  | 22.9  | 20.8  |      | (25) |
|      |                                                                   | 10% MCWB    | 16.0  | 18.3  | 18.1  | 19.4  | 22.2  | 24.5  | 24.9  | 24.3  | 23.7  | 20.8  | 19.3  | 18.9  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 21.0  | 22.0  | 22.3  | 24.3  | 25.2  | 27.4  | 27.5  | 28.1  | 27.3  | 26.4  | 23.6  | 23.0  |      | (27) |
|      |                                                                   | 0.4% MCDB   | 22.1  | 23.1  | 24.0  | 26.8  | 27.9  | 31.0  | 30.3  | 31.2  | 29.0  | 27.9  | 25.4  | 24.1  |      | (28) |
|      |                                                                   | 2% WB       | 19.6  | 21.1  | 21.3  | 23.1  | 24.6  | 26.8  | 27.1  | 27.4  | 26.2  | 25.3  | 22.6  | 21.7  |      | (29) |
|      |                                                                   | 2% MCDB     | 21.1  | 22.6  | 23.6  | 25.7  | 28.2  | 30.6  | 30.1  | 30.5  | 28.6  | 27.5  | 24.8  | 23.2  |      | (30) |
|      |                                                                   | 5% WB       | 18.2  | 20.0  | 20.5  | 22.4  | 24.0  | 26.3  | 26.6  | 26.9  | 25.5  | 24.2  | 21.5  | 20.6  |      | (31) |
|      |                                                                   | 5% MCDB     | 19.8  | 21.9  | 23.2  | 25.1  | 28.1  | 30.2  | 29.8  | 29.8  | 28.6  | 27.3  | 23.9  | 22.0  |      | (32) |
|      |                                                                   | 10% WB      | 16.8  | 18.6  | 19.7  | 21.6  | 23.4  | 25.8  | 26.1  | 26.2  | 24.9  | 23.1  | 20.2  | 19.0  |      | (33) |
|      |                                                                   | 10% MCDB    | 18.2  | 21.0  | 22.3  | 24.4  | 27.3  | 29.7  | 29.6  | 29.1  | 28.4  | 26.0  | 22.5  | 20.5  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.0  | 10.7  | 11.2  | 11.4  | 10.8  | 10.5  | 9.7   | 10.1  | 10.4  | 12.4  | 11.6  | 10.3  |      | (35) |
|      |                                                                   | 5% DB MCDBR | 11.7  | 11.1  | 11.2  | 11.6  | 11.3  | 11.7  | 11.2  | 11.7  | 11.5  | 12.2  | 11.6  | 10.4  |      | (36) |
|      |                                                                   | 5% DB MCWBR | 7.9   | 6.7   | 5.6   | 5.2   | 4.1   | 3.8   | 3.3   | 3.3   | 3.9   | 5.1   | 6.8   | 7.2   |      | (37) |
|      |                                                                   | 5% WB MCDBR | 9.6   | 9.5   | 9.0   | 9.2   | 9.7   | 10.3  | 9.8   | 9.3   | 9.2   | 9.7   | 9.8   | 9.4   |      | (38) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.332 | 0.350 | 0.372 | 0.413 | 0.447 | 0.498 | 0.494 | 0.486 | 0.447 | 0.374 | 0.361 | 0.345 |      | (40) |
|      |                                                                   | taud        | 2.521 | 2.476 | 2.411 | 2.301 | 2.244 | 2.134 | 2.193 | 2.211 | 2.289 | 2.469 | 2.454 | 2.485 |      | (41) |
|      |                                                                   | Ebn at Noon | 899   | 912   | 912   | 881   | 849   | 802   | 804   | 806   | 826   | 873   | 855   | 862   |      | (42) |
|      |                                                                   | Edh at Noon | 87    | 100   | 114   | 132   | 140   | 156   | 146   | 142   | 127   | 99    | 93    | 87    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.80  | 3.15  | 4.39  | 5.34  | 5.93  | 5.98  | 5.92  | 5.62  | 4.97  | 4.26  | 3.20  | 2.45  |      | (44) |
|      |                                                                   | RadStd      | 0.27  | 0.53  | 0.38  | 0.39  | 0.41  | 0.47  | 0.46  | 0.52  | 0.47  | 0.57  | 0.32  | 0.28  |      | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (25 neighbors) | +0.31 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +110    | +79     |

Nomenclature: See separate page